

Snapdragon 8 Elite Hexagon v79 HTP & GenieX Runtime

1. Unified GenieX Execution Pipeline

Snapdragon 8 Elite utilizes Qualcomm's GenieX runtime (`qualcomm/GenieX`) with native GGML Hexagon backend integration.

- ****Direct GGUF Execution****: Any HuggingFace GGUF (e.g. Qwen 3.5 0.8B / 9B Q4_0) executes directly on the NPU without offline host compilation.
- ****Execution Modes****:
 - `--device hybrid`: Per-tensor scheduling offloading supported ops to HTP while falling back gracefully (~90 tok/s prefill, ~27 tok/s decode).
 - `--device npu`: Pinned deterministic execution on HTP0.

2. Low Memory Killer (LMK) Isolation

By running model inference inside isolated native daemons (`:qairt\_engine`, `:llamacpp\_engine`) communicating via UNIX domain sockets, the system avoids Android UI GC pauses and insulates background tasks from game/UI stutters.